Psychiatric Drug Satisfaction: NLP Analysis of Drug Reviews by Diagnosis

Psychiatric Drug Satisfaction

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ABSTRACT

Psychiatric medication adherence remains a significant challenge in mental health treatment, with many patients expressing dissatisfaction with prescribed therapies. This study applies natural language processing (NLP) techniques to analyze over 60,000 user-generated reviews from WebMD, aiming to identify trends in sentiment, common adverse effects, and patient-reported concerns across psychiatric medications. Sentiment analysis using VADER revealed that, despite a slightly higher overall rate of positive sentiment, the most commonly reviewed conditions and medications were predominantly associated with negative sentiment. Named Entity Recognition (NER) using a fine-tuned SciBERT model identified depression, anxiety, and weight-related terms as the most frequently mentioned effects, often appearing in reviews regardless of the prescribed medication. Aspect mining further reinforced these findings, with side effects consistently emerging as the most salient topics for patients. Further analysis of review content also revealed that discussion of symptoms such as depression peaked during periods of societal stress, including the 2008 financial crisis and the COVID-19 pandemic. Together, these findings suggest a disconnect between clinical prescribing practices and patient-reported experiences. By integrating sentiment, entity, and aspect-level analyses, this study demonstrates the potential of NLP to surface actionable insights that can inform more patient-centered psychiatric treatment strategies, improving medication compliance and satisfaction.

1 INTRODUCTION

Understanding how patients with different conditions perceive and react to their medications is essential for improving individualized treatment strategies. Self-reported information may provide us with valuable insight to improve selection of and adherence with proper medications. It is known that medication nonadherence contributes to reduced health outcomes and higher healthcare costs, as well as concerns such as relapse of other substance abuse, so by focusing on this improvement, we better the quality of life of those affected and reduce expenses (Lam & Fresco, 2015); (Marrero et al., 2020). Psychiatric drug adherence is a complex issue with many contributing factors, such as sociological, demographic, diagnostic, and mindset (De las Cuevas & de Leon, 2020); (De las Cuevas et al., 2017).

I have chosen this topic because of my long-standing interest in mental health and personalized medicine. Drug development, particularly in the field of psychiatry, is a vast area for opportunity and growth. Because the mechanisms of many psychiatric drugs are still poorly understood, we should be taking advantage of all the data we have at our disposal to predict the most beneficial medications and save patients from the potential of years of trial and error (Stern et al., 2018). Furthermore, to save them from the harm, both physical and psychological, that is done by experiencing side effects. Precision medicine can and will provide patients with the solace that they are being treated appropriately, and my goal is to identify the areas where that research is best suited, but, while those discoveries are still some time away, there is immeasurable value in providing patients proper treatment with the medications we have currently available to us. As mental disorders affect 1 in 8 people worldwide, there is a vast number of patients that will benefit from improved prescribing and drug development (WHO, 2022).

1.1 Goals

This project aimed to determine whether patients with different psychiatric conditions or on different medications express distinct sentiments when reviewing their medications. I assessed correlations between sentiment, side effects, and common aspects of reviews. Additionally, I identified further trends as they may point to useful information in diagnostics and treatment.

Natural Language Processing (NLP) provides an opportunity to analyze large-scale patient-reported experiences through online reviews, offering insights into sentiment trends across different mental health conditions. The three NLP methods this project uses to achieve the goals are: 1) Sentiment Analysis; 2) Named Entity Recognition (NER); and 3) Aspect Mining (Batutin, 2024).

NER is useful in identifying key terminology like side effects or mentions of other drugs in the free-text field. This method focused on the drug name along with the free-text variables of the dataset. Additional analysis incorporated numerical satisfaction ratings. NER results are helpful in identifying frequent side effects and used in conjunction with other analyses to identify effects are correlated with worse satisfaction or sentiment.

By applying sentiment analysis, this research explored differences in patient sentiment and identified common themes in medication experiences across conditions. This technique made use of the free-text, condition, drug, and numerical ratings variables. The goal was to identify if there were trends of stronger negative sentiment among specific conditions overall, to see if specific drugs have strong negative sentiment, and if more negative sentiment was correlated with poorer numerical ratings. Additional analysis included evaluating if the presence of an effect listed in the NER analysis was correlated with more positive or negative sentiment.

Finally, aspect mining was used to categorize the free-text field to provide further insight into what aspects are most prevalent in reviews. This technique again made use of the free-text and drug variables. These results can help pinpoint the most important factors to patients. Additional analysis also made use of the date of review to evaluate specific aspects prevalence over time. The results will help drive conversations with healthcare providers to get to root causes of dissatisfaction and assist in improving the likelihood of identifying medications which are safe, effective, and that patients are able to adhere to.

2 METHODOLOGY

2.1 Dataset

The CSV dataset was retrieved from Kaggle, created by Parhami (2024). The dataset is sourced from WebMD and contains over 60,000 user-generated reviews from 2007 through 2024. The data contains 12 fields, including structured fields such as the patient's condition, medication name, numerical ratings for effectiveness and satisfaction, and unstructured free-text reviews. This data is a collection of user reviews posted directly to WebMD's public drug database.

2.2 Sentiment Analysis

2.2.1 Data Preprocessing

The preprocessing included some removal of unnecessary rows followed by cleaning steps. I first removed any rows of data that did not have a response in the free-text field. Additionally, I removed rows where the 'Condition' was left blank or written in as "Other" as those are not beneficial for my analysis. Next, I defined a preprocessing function to lower case all text, remove punctuation and numbers using regex, tokenize, lemmatize, and remove stop words (Yan, 2024). Finally, I simplified the drug name column to only the first word as much of what followed was the route of administration or dosage information which inflated the number of unique drugs in the dataset.

2.2.2 Application of the Model

I then used my cleaned dataset to perform VADER sentiment analysis to classify the free text as positive, negative, or neutral. The VADER tool was defined to categorize sentiment calculations of greater than 0.05 as positive and less than -0.05 as negative and saved this analysis to new CSV files. The VADER (Valence Aware Dictionary and sEntiment Reasoner) lexicon-based sentiment tool was selected due to its specific function in identifying sentiment in social media posts (Hutto, 2014). Being that this test is from website reviews, I felt that it was categorically similar to social media, and therefore a good fit.

2.2.3 Statistical Analysis and Visualizations

Further steps included calculating the average scores of the four numerical ratings from the dataset (rating_overall, rating_effectiveness, rating_ease_of_use, and rating_satisfaction) by sentiment category. Then calculated average rating_overall categorized by sentiment and condition, followed by the rating_overall categorized by sentiment and drug. All three of these were again saved to new CSV files.

To generate visualizations, I simplified the number of values to be presented. I filtered for the top 15 conditions and drugs based on the number of total reviews received. I then used those to generate two stacked bar graphs, Sentiment Analysis by Condition and Sentiment Analysis by Drug. I also generated a histogram of the four numerical ratings averages grouped by sentiment.

2.3 Named Entity Recognition

2.3.1 Data Preprocessing

To preprocess my dataset for NER, I first repeated the step done for sentiment analysis of generating a new column with just the first word of the drug name so it was available in this new data frame. I then also dropped rows with empty free text fields and truncated the free text to 510 tokens, due to the model adding two hidden tokens when processing, keeping it to the 512 max BERT allows.

2.3.2 Application of the Model

I loaded a sciBERT NER model from Lee (2021), on Huggin Face, that was designed to identify adverse effects and drug names from free text. Prior to running my processed data, I ran a sample text to ensure the model performed in the manner expected. I used a singular text field from a review and displayed the results using displacy.render() to confirm I get my desired outcome. I then ran my preprocessed data though the model and saved it to its own data frame and CSV download.

2.3.3 Statistical Analysis and Visualizations

I viewed a count read out of the top 10 most common effects identified by the model. I followed that with a bar plot of that data. I then calculated the top 3 most common effects for each of the top 10 most common drugs and plotted that as a bar plot. I additionally wanted to see if the number of effects identified was correlated to the number of reviews a drug got. This was visualized with a scatter plot.

2.4 Aspect Mining

2.4.1 Data Preprocessing

The preprocessing for the aspect mining model was the same as the sentiment analysis, including removal of unnecessary rows of data that did not have a response in the free-text field. Additionally, I removed rows where the ‘Condition’ was left blank or written in as “Other.” Next, I defined a preprocessing function to lower case all text, remove punctuation and numbers using regex, tokenize, lemmatize, and remove stop words (Yan, 2024). I also simplified the drug name column to only the first word.

2.4.2 Application of the Model

I then used my cleaned dataset to perform aspect mining using the spaCy en_core_web_sm model, a rule-based approach. I then extracted and listed the top 10 identified aspects. The first several runs of this model provided mostly generic or non-useful terms like pronouns or drug names, so I generated a list of these terms to be excluded in the aspect extraction. When I was satisfied with the results, I exported the results to a CSV file.

2.4.3 Statistical Analysis and Visualizations

Using the top 10 aspects, I generated a simple bar chart. I then generated a bar plot similar to the NER model where I plotted the top 3 aspects for each of the top 10 medications. I then evaluated the prevalence of several aspects over time, including "depression," "pain," and "weight," and plotted each as a line plot.

2.5 Further Analysis

Following that, I generated a merged data frame, exported to CSV, and generated visuals of the average satisfaction scores grouped by the presence or absence of effects identified by the NER model, and a bar plot of the presence of effects grouped by sentiment.

2.6 Tools & Libraries Used

Python libraries used include pandas, matplotlib, seaborns, nltk, regex, spaCy, collections, datasets, transformers, and google.colab. The computing environments were Google Colab and Juptyer Notebook.

3 RESULTS

3.1 Sentiment Analysis

The goal of the sentiment analysis was to determine if there were any relationships between condition, drug, ratings, and the sentiment of the free-text review. Of over 47,000 reviews that met criteria to undergo sentiment analysis 50.3% showed positive sentiment, 9.0% were neutral, and 40.6% were negative as seen below (a).

	vader_sentiment	avg_rating_overall	avg_rating_effectiveness	avg_rating_ease_of_use	avg_rating_satisfaction	count
0	negative	3.415265	3.329238	3.981906	2.936484	23931
1	neutral	3.528833	3.388077	4.018093	3.179773	4311
2	positive	4.030159	3.977486	4.405000	3.707469	19321

Figure a. Average ratings by sentiment

The breakdown of (b) Sentiment Analysis by Condition and (c) Sentiment Analysis by Drug of the top 15 in their category can be seen below. Additionally, is the histogram (d) of numerical ratings averages grouped by sentiment.

As seen in figure b, depression is the most common condition resulting in reviews. Additionally, it does show over 50% negative sentiment, as do the other conditions in the top 15. This is significant because the overall dataset showed over 50% positive sentiment analysis. Figure c also shows near to over 50% negative sentiment on the top 15 percent of drugs. Figure d depicts that there is a slight upward trend in scoring as we shift from negative to positive sentiment, with ease of use being consistently the highest rated.

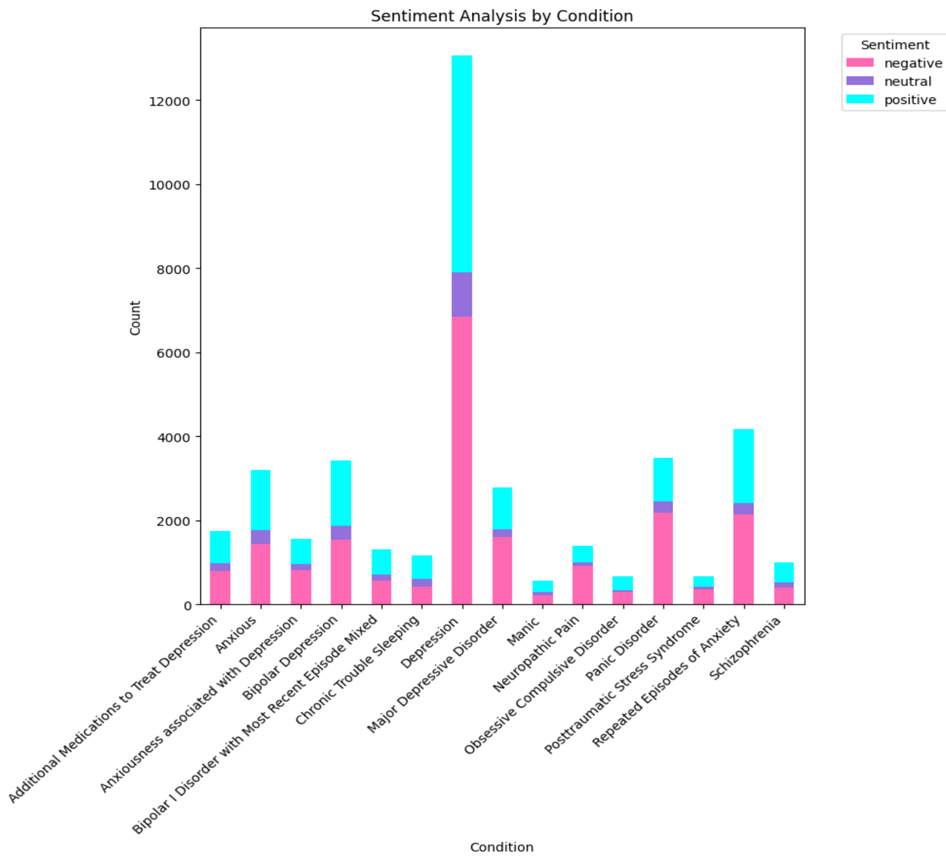


Figure b. Sentiment analysis by condition. Depicted as a stacked bar chart.

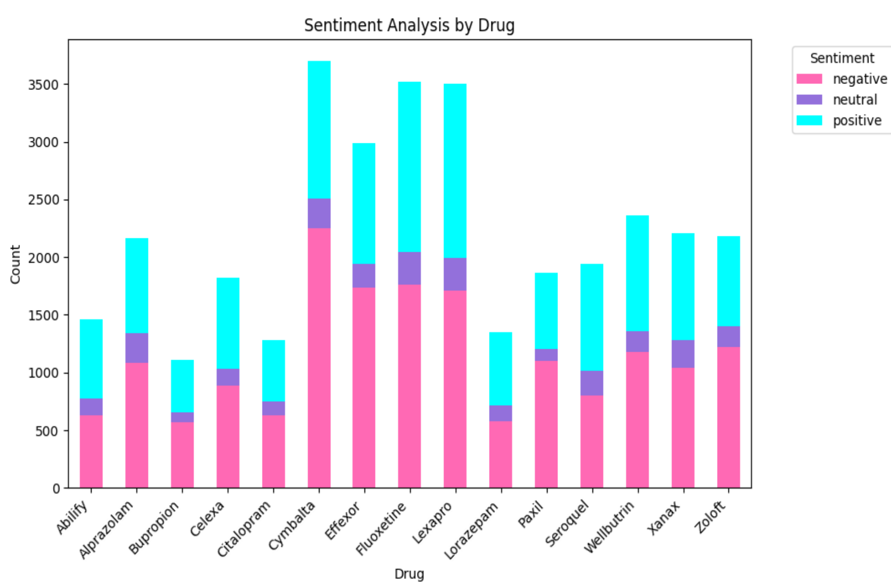


Figure c. Sentiment analysis by drug. Depicted as a stacked bar chart.

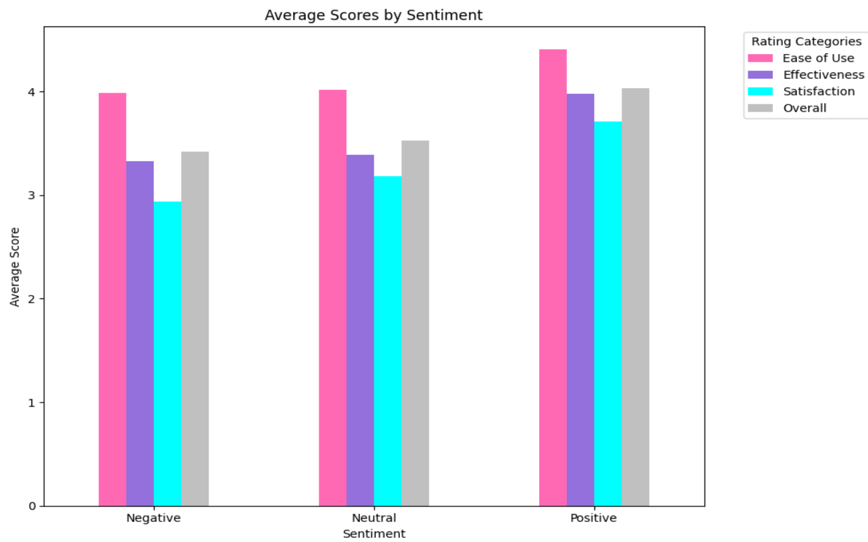


Figure d. Average sentiment analysis scores. Depicted as a histogram.

3.2 Named Entity Recognition

The Named Entity Recognition (NER) model was used to identify frequently mentioned side effects and symptoms in user-generated reviews. As shown in Table 1 and Figure e, the 10 most common effects included “depression,” “anxiety,” and “weight gain,” with “depression” appearing in over 4,300 reviews. These terms were not only frequent but also aligned with common psychiatric symptoms, raising the possibility that patients are using medication reviews to express both treatment outcomes and underlying conditions.

Effect	Count
depression	4331
anxiety	3153
panic attacks	2462
weight gain	2428
nausea	1431
headaches	1390
tired	1229
weight	1199
dizziness	1022
insomnia	980

Table 1. Top 10 identified adverse effects by NER model.

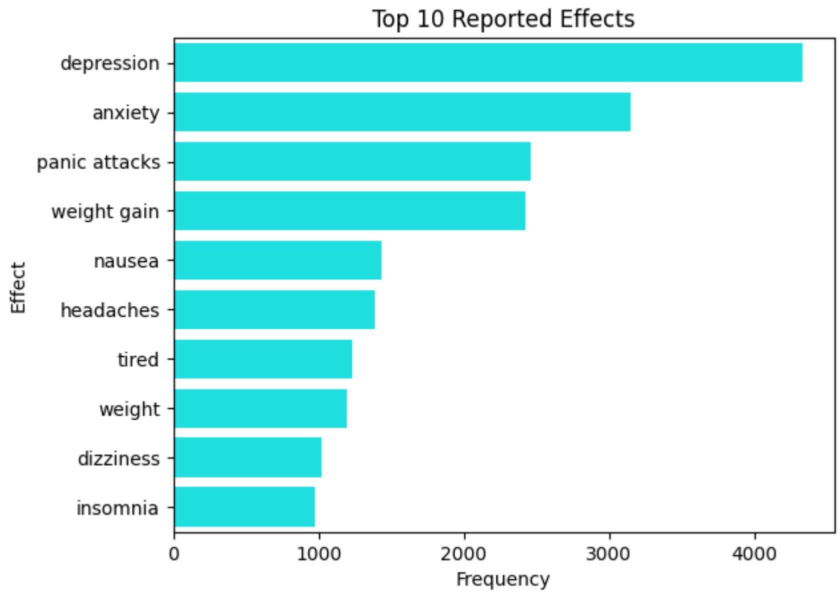


Figure e. Top 10 identified adverse effects by NER model. Depicted as a bar plot.

Figure f presents the top three effects identified per each of the 10 most reviewed drugs. “Depression” appeared in 9 of the 10 drug profiles, underscoring its prominence in patient discourse. Interestingly, while “pain” was one of the least frequently mentioned effects overall, it still appeared in at least 1 of the most-reviewed drugs, suggesting that specific side effects may be tied to only certain prescriptions.

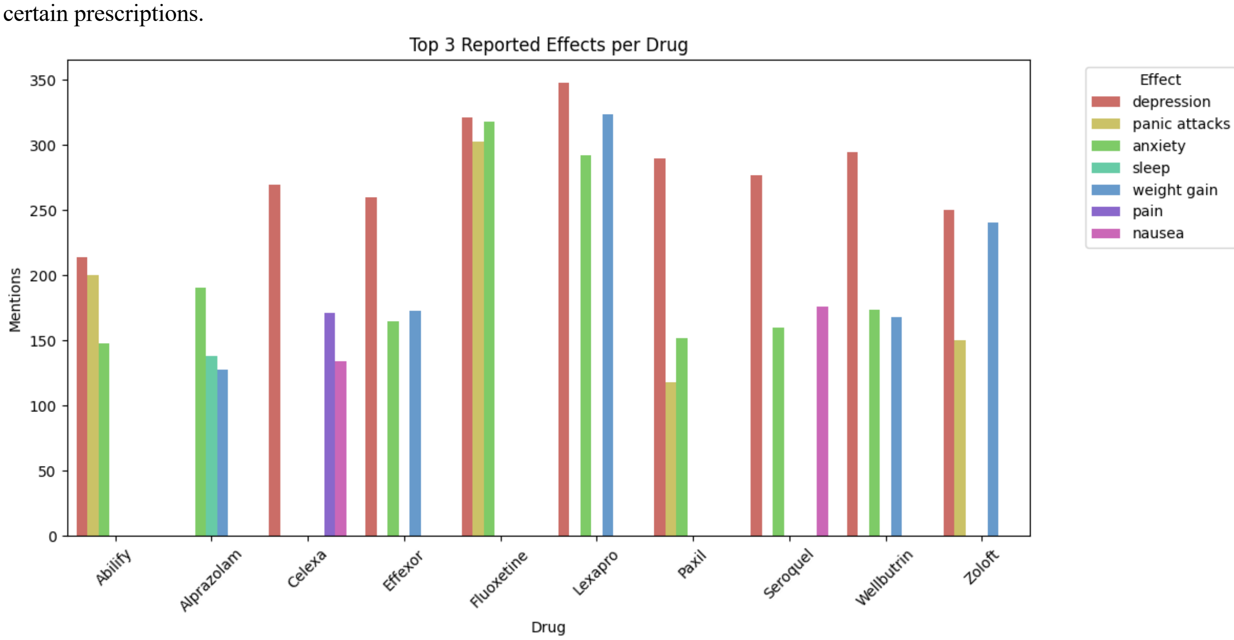


Figure f. Top 3 Adverse Effects per Top 10 Drugs. Depicted as a bar plot.

A further analysis explored the relationship between the number of reviews and the volume of effects identified (Figure g). The scatter plot demonstrated a near-linear relationship between review volume and number of effects mentioned, implying that more reviews provide a proportional representation of user-reported effects. An outlier was identified for the drug Cymbalta, which had the highest number of reviews (4,690) but significantly fewer effect mentions (1,740). This suggests that, while widely prescribed, Cymbalta may provoke less explicit mention of adverse effects, or that users may focus on other dimensions of the experience such as efficacy or cost, which could be further explored in future studies.

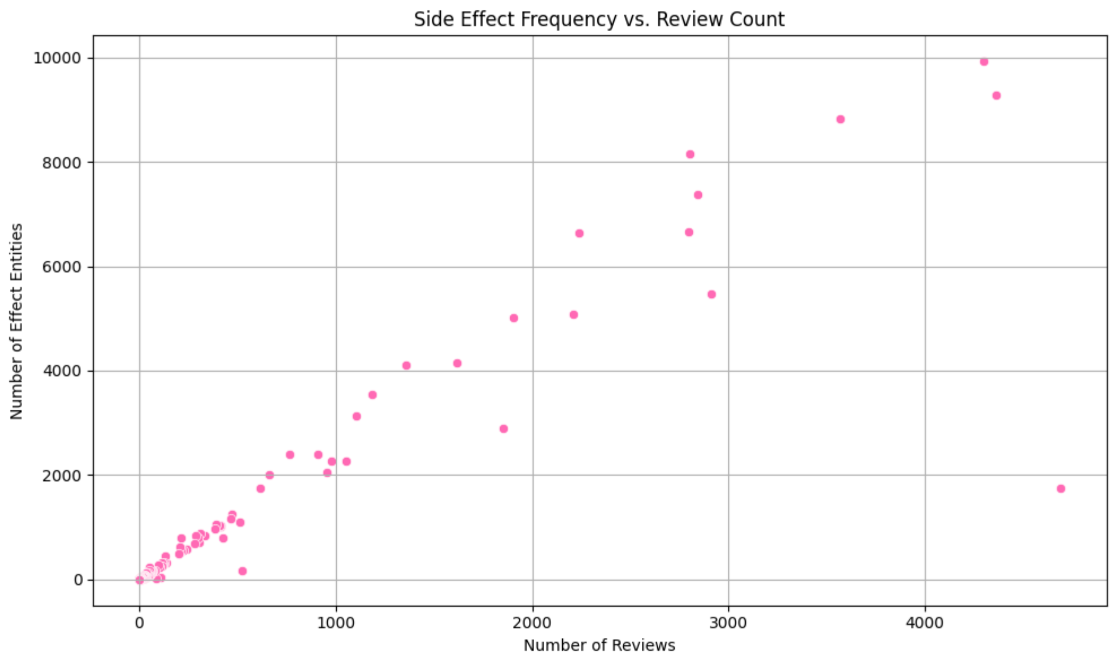


Figure g. Number of Adverse Effects vs Review Count. Depicted as a scatter plot.

3.3 Aspect Mining

The aspect mining analysis aimed to surface the most frequently discussed concerns in patient reviews. As shown in Figure h, the top 10 aspects extracted from the text predominantly relate to adverse effects, with "depression" and "anxiety" emerging as the most prominent themes. Notably, there was strong overlap between these aspects and the entities identified by the NER model, reinforcing the centrality of emotional and mental side effects in user experience.

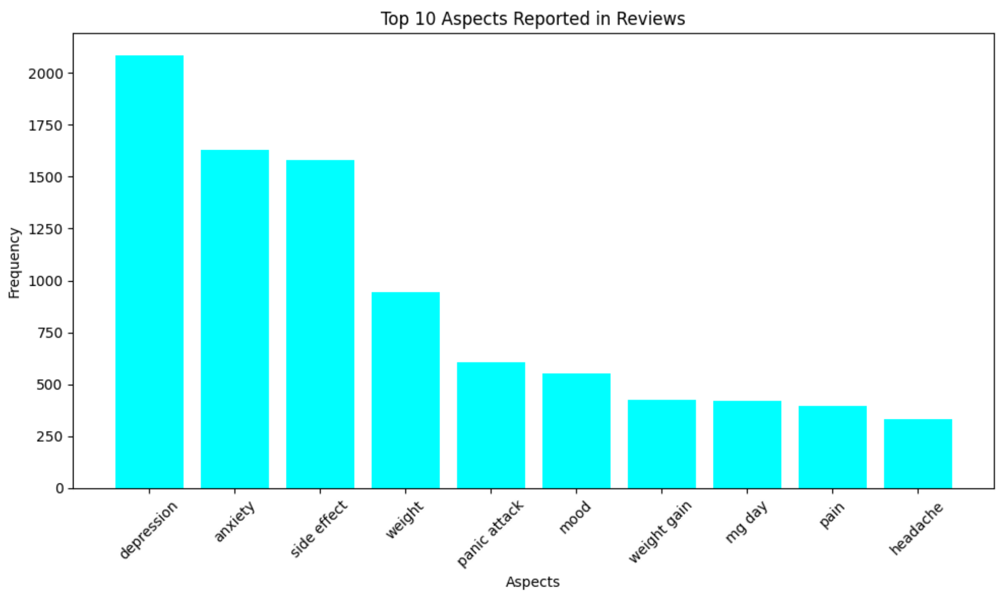


Figure h. Top 10 Aspects Identified in Reviews. Depicted as a bar plot.

Figure i further illustrates the top 3 aspects associated with each of the 10 most reviewed medications. “Depression” and “anxiety” appear across 8 of the 10 drugs, suggesting that these concerns are both prevalent and generalizable, regardless of the medication prescribed. This consistency highlights a recurring theme of dissatisfaction in psychiatric drug treatment and suggests that future drug development or guidelines should place greater emphasis on managing these psychological side effects.

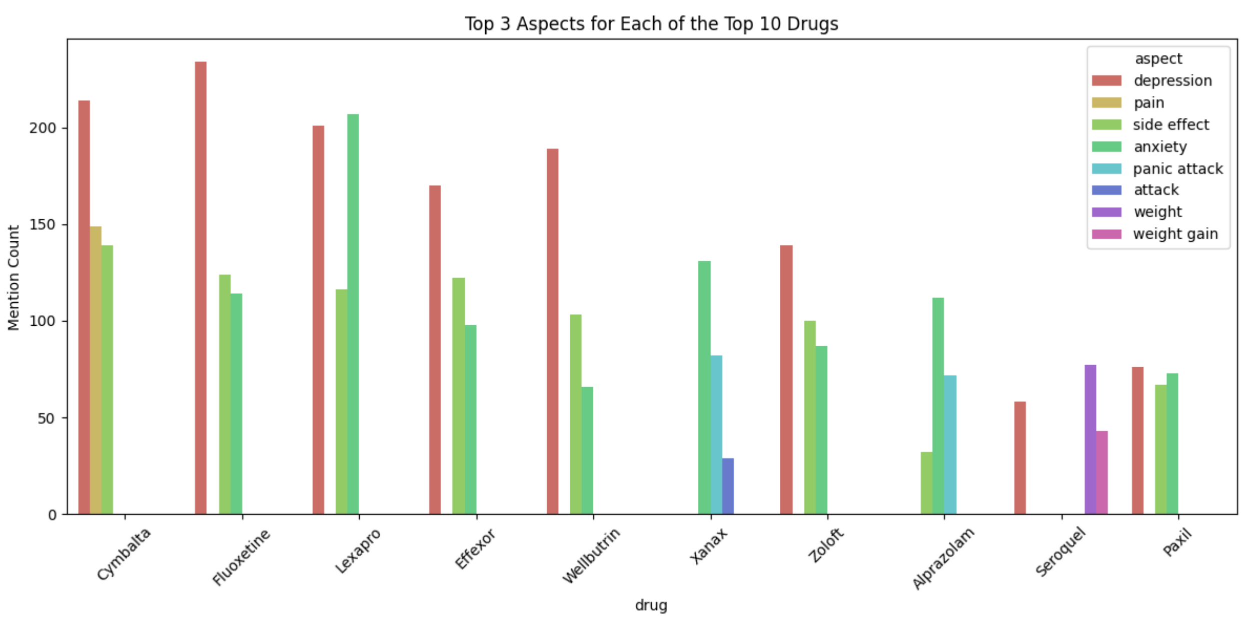
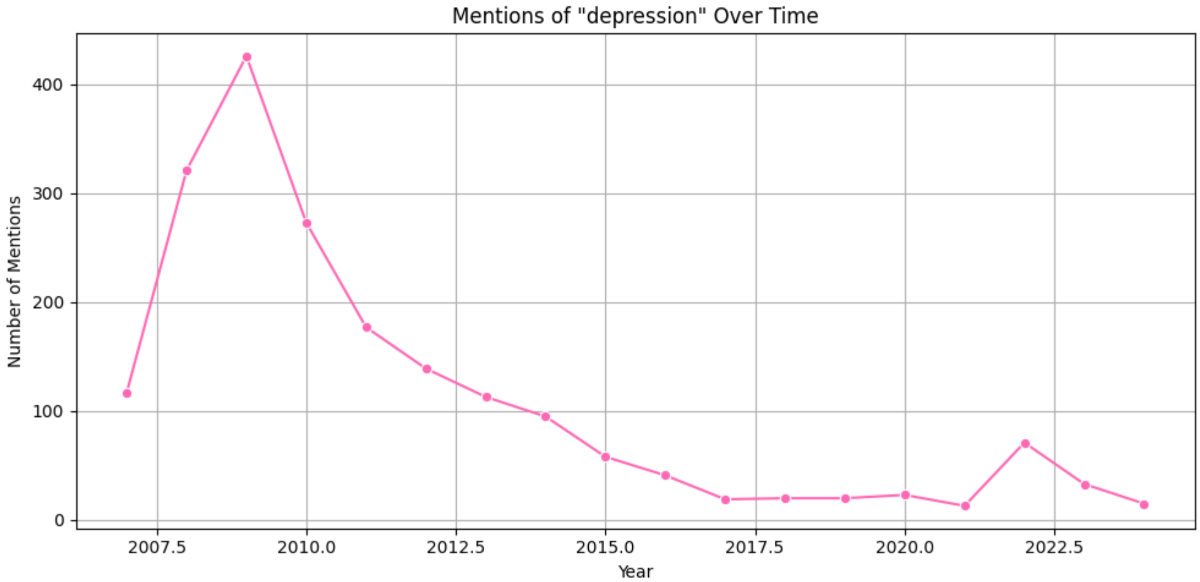
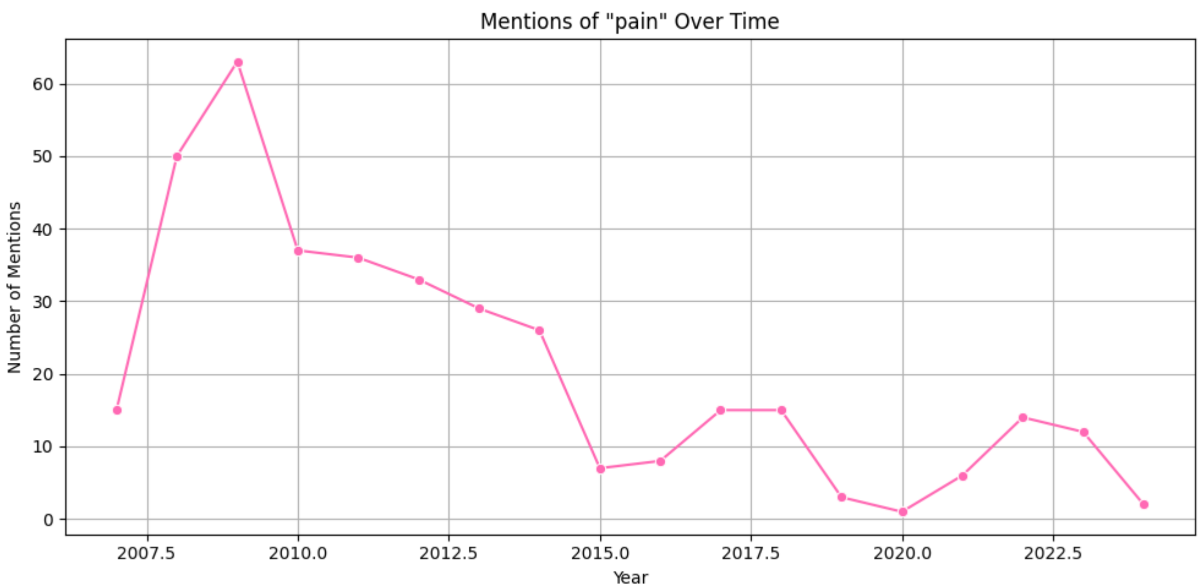


Figure i. Top 3 Aspects per Top 10 Drugs. Depicted as a bar plot.

Additional analysis of aspect mentions (Figure j) showed that terms like “depression,” “pain,” and “weight” followed a similar trajectory, peaking around 2008–2009 and again in 2021. These patterns may correspond to known societal stressors, such as the global financial crisis of 2008 and the COVID-19 pandemic, and align with prior studies linking economic downturns to increased psychiatric distress (Frasquilho et al., 2016). These insights suggest that reviews can be influenced not only by drug efficacy but also by external psychosocial factors, further complicating adherence and satisfaction trends.





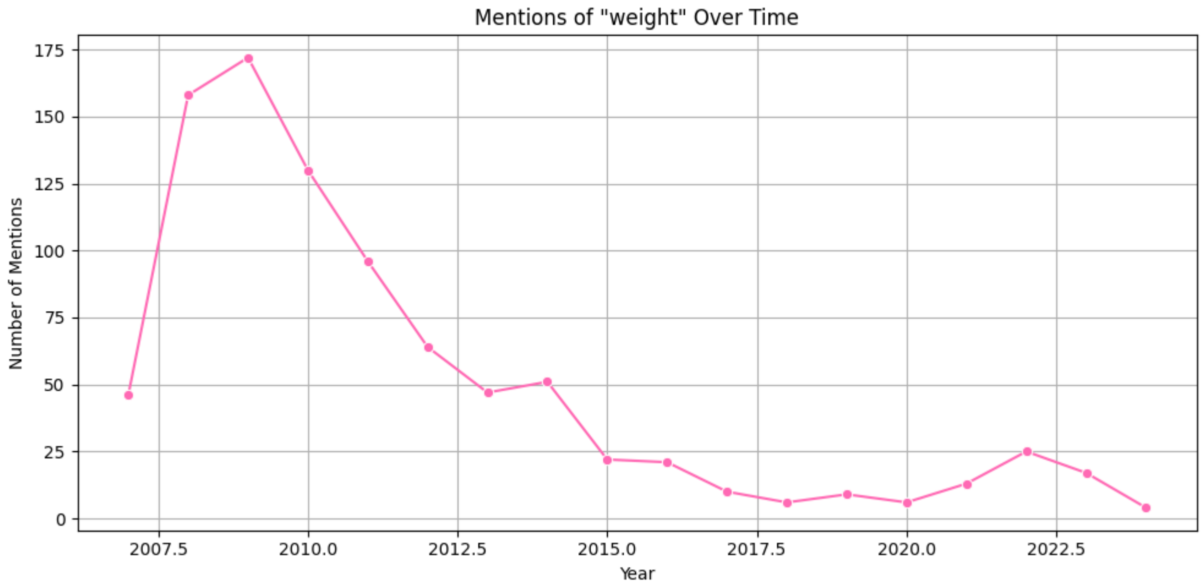


Figure j. Mentions of Depression (top), Pain (middle), and Weight (bottom) over time. Depicted as line plots.

3.4 Further Analysis

To explore connections between NLP outputs, a merged data frame was created that combined results from sentiment analysis, NER, and aspect mining. The first analysis compared average satisfaction scores between reviews that did and did not contain recognized effect entities. As shown in Figure k, the difference was minimal: reviews with no recognized effects averaged a satisfaction score of 3.33, while those with at least one effect averaged 3.24. This finding suggests that, while adverse effects are a dominant review theme, their mere presence does not necessarily predict overall satisfaction. Other factors, such as treatment outcome expectations or external stressors, may influence ratings more significantly.

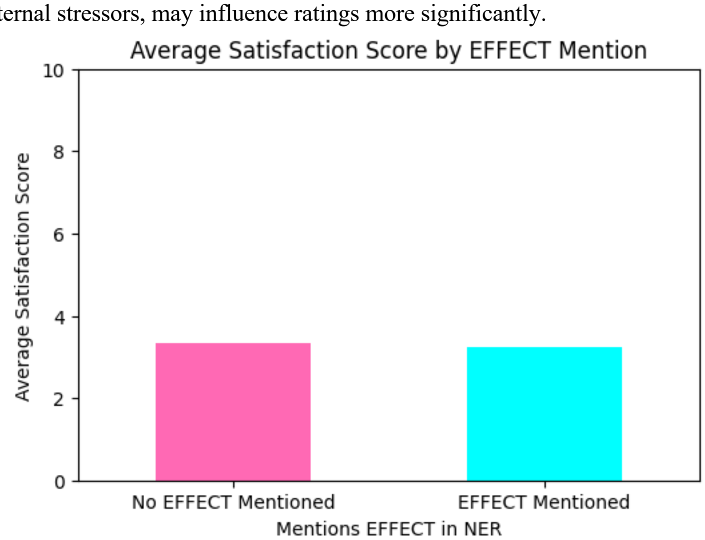


Figure k. Average Satisfaction Score by Presence of Effects in Review

The second combined analysis (Table 2 and Figure l) investigated the frequency of effect mentions across sentiment categories. Results showed that reviews with negative sentiment and effect mentions were the most common, comprising 14,877 reviews. Interestingly, a significant portion of positive reviews (11,873) also mentioned effects. This suggests that some patients are able to tolerate or contextualize side effects within an overall positive treatment experience, a nuance that would be missed by sentiment or entity recognition alone. The strong overlap between negative sentiment and mentions of adverse effects highlights how NER can help identify specific sources of patient dissatisfaction. At the same time, the presence of effect-related terms in many positive reviews suggests that side effects alone do not determine satisfaction, underscoring the importance of interpreting patient feedback in context rather than in isolation.

Sentiment	Effect	Count
negative	No EFFECT	3883
negative	Has EFFECT	14877
neutral	No EFFECT	751
neutral	Has EFFECT	2624
positive	No EFFECT	3216
positive	Has EFFECT	11873

Table 2. Presence of Effects in Reviews by Sentiment Label.

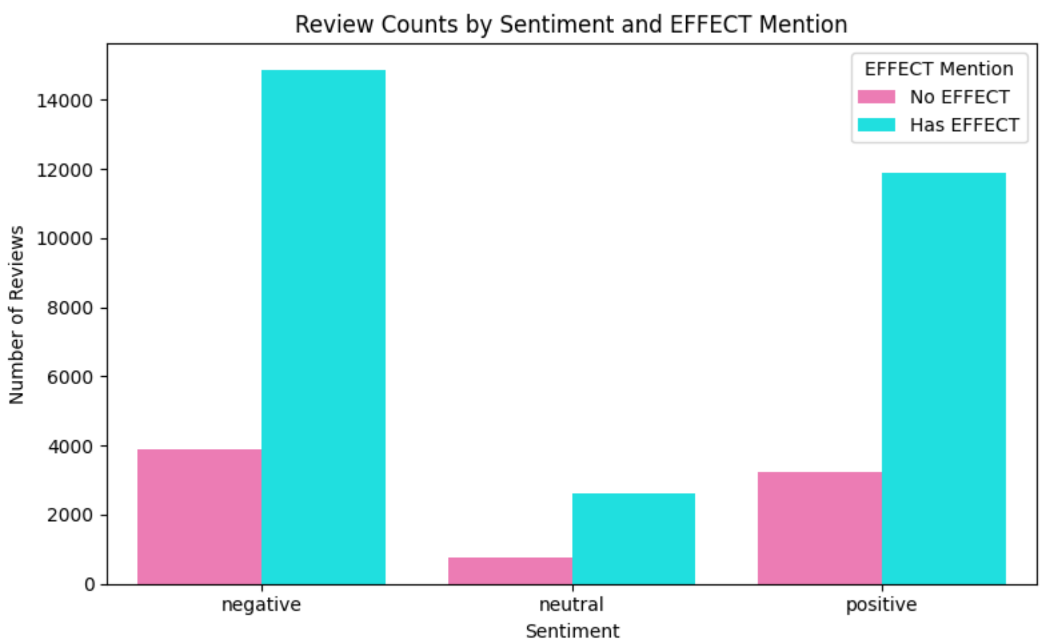


Figure l. Presence of Effects in Reviews by Sentiment Label. Depicted as a bar plot.

4 DISCUSSION

4.1 Sentiment Analysis

Though the overall dataset showed over 50% positive sentiment in reviews, all 15 of the top treated conditions showed over 50% negative sentiment. Similarly, the top 15 most reviewed drugs also all hovered around or above 50% negative sentiment.

This may be suggestive that the top 15 conditions are not being treated with the most appropriate drugs, but due to the commonality of these conditions, providers prescribe medications based on habit and not individual needs. Likewise, the top 15 drugs may be so commonly prescribed that providers are not taking the time to determine if these are the best treatments for these patients. It may also be suggestive that these drugs are causing unintended reactions or side effects in patients as “Satisfaction” is consistently rated the lowest.

As for a goal of this sentiment analysis being to determine if any conditions constantly rate drugs to be worse than others, it is evident that due to the multiple variations of depression and anxiety appearing in the top 15, most with notable negative sentiment score, it does appear that there is a correlation between diagnosis and sentiment of the review.

Drug sentiment analysis appears to hold more around 50% negative but notably the SSRIs and SNRIs consistently appear to have the most negative sentiment. This is consistent with literature showing that both SSRIs and SNRIs show overall more negative sentiment, with SNRIs lower on average (Compagner et al., 2021).

The limitations of this analysis include that of VADER sentiment analysis. As this tool was designed for social media, there may be some errors in sentiment calculations of medication reviews. Additionally, the dataset itself may have mild biases, as those who leave public reviews may not be indicative of a representative diverse sample. Finally, some errors may be introduced in the preprocessing steps. As having reviews that did not provide a free-text portion, or marking "Condition" as other or blank, resulted in approximately 15,000 of the original 60,000 reviews being removed.

4.2 Named Entity Recognition

Named Entity Recognition results indicate that depression and anxiety are commonly discussed effects by reviewers. For example, the top 3 recognized effects in reviews for Alprazolam were anxiety, sleep, and weight gain. Alprazolam is a medication that treats anxiety with known side effects of tiredness and weight changes, so it is expected to see those as the 3 top noted effects by the NER model (American Society of Health-System Pharmacists, 2021). However, for something like Seroquel, used to treat schizophrenia and bipolar disorder, depression, anxiety, and nausea were the top 3 effects. Though Seroquel has a red box warning indicating the potential changes in mental health when increasing or decreasing a dose, depression, anxiety, nor nausea are listed as known side effects of this drug (American Society of Health-System Pharmacists, 2020). This indicates that the drug may benefit from additional studies to identify unrecognized side effects and prompt conversations with providers to have with patients recently started on this medication. Additionally, it was found the drug with the most reviews, Cymbalta, had proportionally far fewer effects identified than other highly reviewed drugs. This may suggest that Cymbalta, a serotonin and norepinephrine reuptake inhibitor (SNRI) antidepressant, may have factors other than side effects contributing to review opinion, as Cymbalta was shown to have a majority of negative sentiment reviews.

While this NER model does successfully identify the various effects in the free text, it is limited to what is pre-defined, in this case effects and drugs. Additionally, it is not able to distinguish between symptoms and adverse effects listed in the free text. Depression and anxiety were the most common effects identified however, 8 of top 10 drugs are also used to treat anxiety and/or depression, which presents a challenge when analyzing the results of the NER model. Future work on this and similar datasets may benefit from a more robust models that can differentiate between intended effect and adverse effect.

4.3 Aspect Mining

Aspect mining was performed to identify the top factors that reviewers find important. To capture the most relevant factors, I filtered out common or unhelpful words that were used frequently, like dosing information or terms like "doctor" or "medicine." What was left was a majority more side effect mentions. This reinforces the idea that what patients find to be most significant is the side effects they experience. With the large overlap of aspects identified, and entities identified by NER, it is clear that the most common side effects are also the most common overall themes of the reviews. This should encourage the field to look toward drug development research aimed at reducing side effects to enhance the patient experience.

Additional evaluation of the aspect mining results included observing the prevalence of specific aspects over time. The reviews span from 2007 through 2024. When plotting the mentions of "depression," "pain," and "weight" over time, they all follow a similar pattern of a peak in 2008 and a decline with a small peak in 2021. Mentions of pain over time are the most varied suggesting less of a direct relationship compared to that of weight and depression. The peaks in the depression graph correlate to economic downturns, the market crash of 2008 and the height of COVID-19. It has been studied and found that economic recession is associated with increased rates of mental health disorders (Frasquilho, et al., 2016). This should highlight to medical and psychiatric care providers to take a proactive stance during economic downturns and to educate their patients on their options and the risks before diagnosis or medication becomes necessary, which may alleviate some of the negative responses to new medications. Aspect mining provides insight into what factors reviewer care most about and write most about, but it may have limitations with identifying high value aspects, which were mitigated with exclusionary words. Aspect mining is also limited in its ability to identify nuanced language, slang, or irony if present in this dataset.

4.4 Further Discussion

It was identified that the average satisfaction score between reviews with mentions of effects entities was nearly the same as those without, both around 3.3 out of 5. This may indicate that reviews tend to trend toward the average with no notable difference in those that did and did not mention side effects. Finally, I reviewed mentions of effects categorized by sentiment score. It was found that majority of each sentiment category consisted of reviews that listed effects and that negative reviews listed effects at a higher rate than neutral or positive.

5 CONCLUSION

This study applied three natural language processing techniques, sentiment analysis, named entity recognition, and aspect mining, to over 60,000 psychiatric medication reviews, with the goal of uncovering patterns in patient satisfaction and identifying common themes in medication experiences. The results consistently demonstrated that negative sentiment dominates the most commonly treated conditions and prescribed medications, raising important concerns about how psychiatric drugs are selected and evaluated by patients.

Named entity recognition revealed that side effects such as depression, anxiety, and weight gain were among the most frequently mentioned entities across the dataset. These effects were prevalent across multiple drug types and often appeared in both negative and positive reviews, suggesting a nuanced relationship between adverse experiences and overall satisfaction. Aspect mining further supported these findings, indicating that side effects were not only commonly reported but were also among the most important themes to patients. Further trends in the data also suggested that external societal pressures, such as economic downturns, may influence the way patients experience and report on their medication use.

Together, these methods offer a more complete picture of patient perspectives, identifying not just how people feel about their medications, but why. The integration of structured and unstructured data, including numerical ratings, free-text reviews, and model-extracted insights, provides a path forward for using patient voices to inform treatment strategies. While there are limitations in the interpretability of some models and the inherent bias in self-reported data, the consistent patterns across all three NLP methods suggest real and actionable concerns in psychiatric drug adherence and satisfaction. Future work in this area should continue to refine these techniques, with an emphasis on differentiating between intended treatment effects and adverse outcomes, and expanding the scope to more diverse and representative datasets.

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